

What is your perspective on the country's current state of EV charger infrastructure?

With almost every major automotive OEM introducing electric cars, buses, and even commercial vehicles, the EV (electric vehicle) charging network in India is evolving. Yet, the critical aspect to consider is the pace of this development relative to the rate of EV adoption. The pace of infrastructure development must match the rate of EV adoption. For instance, only about 1% of monthly four-wheeler sales are currently electric vehicles. A significant shift will likely occur once this number reaches a threshold, say 5% or 7%. This inflection point will likely spur charge point operators (CPOs) to accelerate the deployment of more charging stations, further supporting the EV transition.

What are the various business models currently employed by charge point operators (CPOs) in India? How do you see these models evolving or transforming in the future?

CPOs in India primarily operate with two business models: B2B (business-to-business) and B2C (business-to-consumer) retail charging. The B2B model presents a clear business opportunity that positively impacts profit and loss statements. In contrast, the B2C model involves demand risk, as it entails setting up chargers in various locations, including highways and urban areas, with the expectation of growing utilisation.

It's important to note that these models are interdependent. The top CPOs in the country, including the top five or seven, have adopted this dual approach for growth and sustainability. Globally, CPOs are generally yet to be profitable. At ChargeZone, we only recently achieved a positive EBITDA (earning before

interest, taxation, depreciation, and amortisation). The business model for CPOs is clear-cut: focusing on delivering energy or EV charging as a service. This requires substantial capital investment—by my estimates, around \$10 billion for both slow AC and fast DC chargers—to facilitate a 30% EV adoption rate in India by 2030. The critical factor here is the pace of infrastructure expansion and the strategic placement of charging stations. In conclusion, while the 'chicken or egg' dilemma is often debated in the context of infrastructure and EV adoption, infrastructure must precede adoption. However, the speed at which this infrastructure is developed is paramount.

Is capex the most significant challenge, or is there a bigger challenge than that?

Multiple challenges are inevitable in any business, especially one as capital-intensive as the EV charging network. Capital is indeed a significant barrier, not because of CPOs but due to the inherent nature of the industry. However, capital is one of many hurdles. A crucial aspect beyond capital is ensuring customers a reliable and seamless charging experience. This is where 'reliability anxiety' comes into play, perhaps more so than 'range anxiety.' Charging stations are becoming more visible across various locations like highways, gas stations, and hotels. Yet, the critical issue is their operational reliability. Since most charging stations are uncrewed, maintaining consistent uptime is a significant challenge for CPOs. Other challenges, though seemingly routine, are also vital. One such aspect is the quality of the location. Given that charging can take 15 to 30 minutes, providing customers with additional facilities is essential to utilise their time

effectively. Catering to the needs and comfort of individual EV owners and fleet drivers is as critical as our charging stations' capital investment and reliability.

What options are available for individuals interested in capitalising on the business opportunity of setting up EV chargers, particularly DC fast chargers?

If you possess a strategic piece of land or an ideal location, we offer partnership models like the DOCO (dealer owned and charger operated) or COCO (charger owned and charger operated) models. We can work with you in these arrangements, investing in your location to set up the charging infrastructure. However, if you, as an individual, are considering setting up a charging station independently, I suggest caution at this stage. It is best if you are ready to take on the demand and even the right-of-way risks. The feasibility of such an investment depends significantly on the amenities available at the location. For instance, if you don't own a restaurant, dhaba, or a facility along the highway, investing in a charging station might be premature. In such cases, the predictability of utilisation and return on investment is still being determined. That said, CPOs like ours constantly seek valuable partnerships based on location. We are open to engaging with individuals who have suitable sites for setting up charging stations.

In the context of charge point operators, do you encounter similar financing challenges in your business as faced by the EV OEMs, or is this issue not prevalent in the CPO sector?

Indeed, access to capital is a significant barrier for any CPO looking to expand, enter new territories, or explore new markets. The financing of EVs is crucial, affecting the